

SOC Incident Investigation Report

Brute-Force Attack Detection & Analysis | Simulated Lab Environment

Incident ID	INC-2026-001	Analyst	SOC L1 Intern
Date	2026-06-20	Severity	HIGH
Tool Used	Splunk Enterprise 10.4	Status	Confirmed — True Positive
MITRE ATT&CK	T1110 — Brute Force	Host/Source	Kali Linux (Lab)

1. Executive Summary

A brute-force login attack was detected against the **admin** account originating from source IP **192.168.1.50**. The Splunk SIEM identified **6 consecutive failed login attempts (EventCode 4625)** within a 29-second window, followed immediately by a **successful login (EventCode 4624)** at **10:01:31**. This pattern is consistent with a successful brute-force attack resulting in unauthorized account access. The incident is classified as a **True Positive — HIGH severity**.

2. Detection Method

Tool: Splunk Enterprise 10.4 — Search Processing Language (SPL)

SPL Query Used — Brute Force Detection

```
index=main sourcetype=csv EventCode=4625
| stats count by user, src_ip
| where count > 3
```

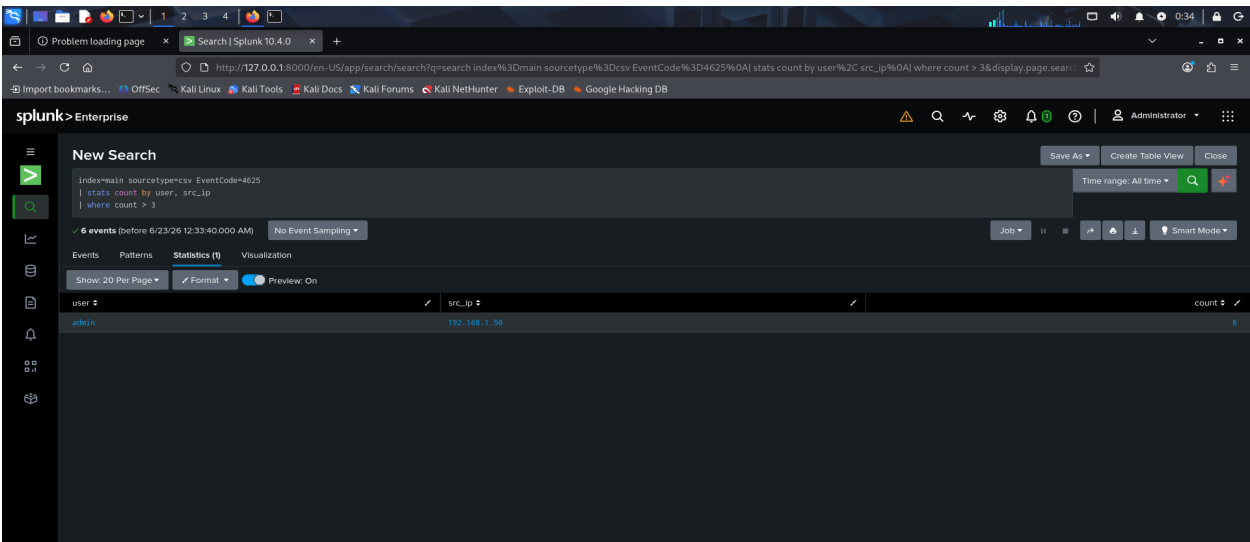


Figure 1: Splunk SPL query result — admin user flagged with 6 failed logins from 192.168.1.50 (threshold: >3)

3. Attack Timeline

The timeline below shows the complete sequence of events for the **admin** account. All events originated from a single source IP, within a 30-second window — a strong indicator of an automated brute-force tool rather than manual login attempts.

Timestamp	EventCode	Status	Description
2026-06-20 10:01:02	4625	FAILURE	Failed login attempt #1
2026-06-20 10:01:05	4625	FAILURE	Failed login attempt #2
2026-06-20 10:01:09	4625	FAILURE	Failed login attempt #3
2026-06-20 10:01:14	4625	FAILURE	Failed login attempt #4
2026-06-20 10:01:20	4625	FAILURE	Failed login attempt #5
2026-06-20 10:01:25	4625	FAILURE	Failed login attempt #6
2026-06-20 10:01:31	4624	SUCCESS	Successful login — ACCOUNT COMPROMISED

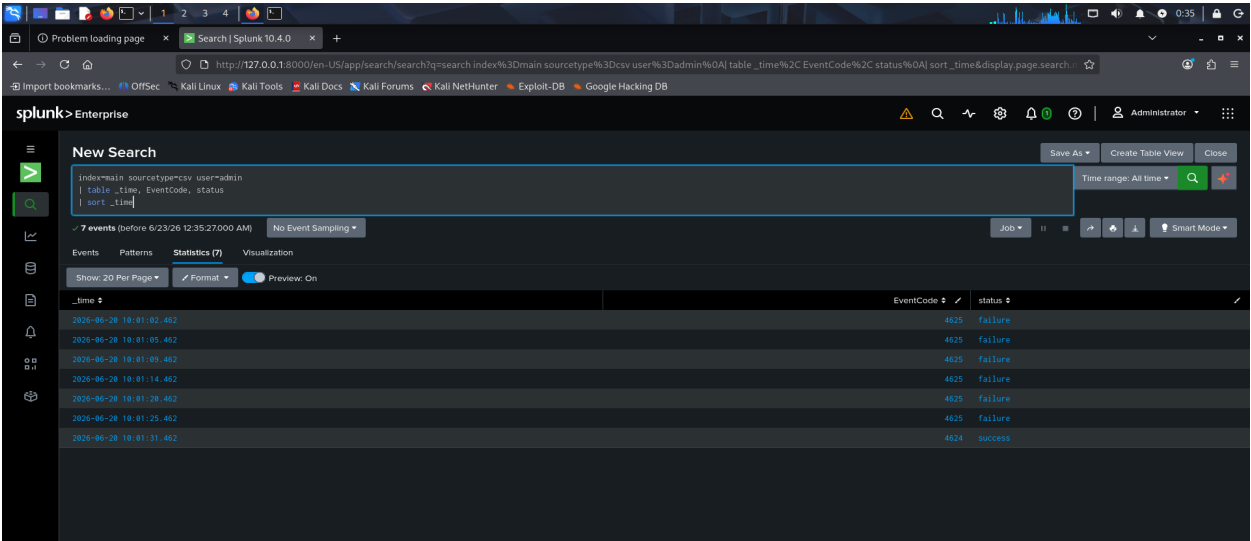


Figure 2: Splunk timeline view — 6 failures (EventCode 4625) immediately followed by successful login (EventCode 4624)

4. Indicator of Compromise (IOC) Analysis

IOC Type	Value	Reputation	Notes
Source IP (Internal)	192.168.1.50	Private IP	RFC1918 private range — internal threat actor or compromised host
Source IP (External)	45.33.22.11	Linode/Akamai — 65 reports (0% abuse rating)	Cloud hosting IP — could be attacker-controlled VPS
Target Account	admin	N/A	High-privilege account — critical asset
EventCode	4625 / 4624	N/A	Windows failed / successful logon events

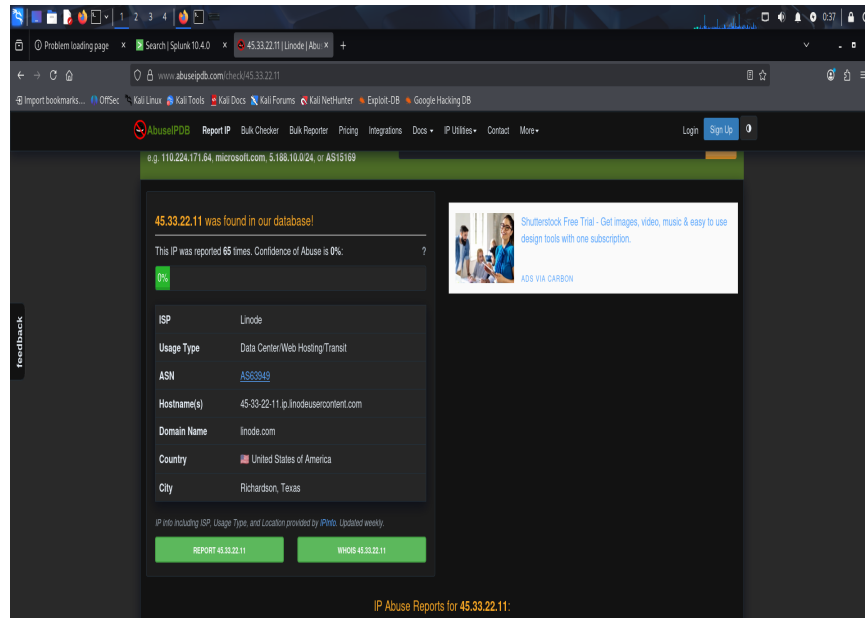


Figure 3: AbuseIPDB result for 45.33.22.11 — Linode hosting IP, 65 reports on record

5. Verdict & Classification

Classification	True Positive
Attack Type	Brute Force Login Attack (Credential Stuffing / Password Spraying)
MITRE ATT&CK	T1110 — Brute Force T1078 — Valid Accounts (post-compromise)
Severity	HIGH — Privileged account (admin) successfully compromised
Confidence	HIGH — Pattern consistent with automated brute-force tool (30-sec window, 6 attempts)

6. Recommended Remediation Actions

Priority	Action	Owner
IMMEDIATE	Reset admin account password and invalidate all active sessions	L2 Analyst / IT Admin
IMMEDIATE	Isolate source host (192.168.1.50) from network pending investigation	L2 Analyst
HIGH	Block external IP 45.33.22.11 at perimeter firewall	Network/Firewall Team
HIGH	Enable account lockout policy (e.g., lock after 5 failed attempts)	IT Admin
HIGH	Enable Multi-Factor Authentication (MFA) on admin account	IT Admin
MEDIUM	Review all activities performed under admin account post-compromise	L2/L3 Analyst
MEDIUM	Add IOCs (IPs, EventCode patterns) to SIEM detection rules	SOC Engineer
LOW	Conduct post-incident review and update runbook	SOC Manager